Miriam Shem

Software Engineer | AI Solutions Engineer | Digital Public Infrastructure

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PROFESSIONAL PROFILE

Software Engineer with professional experience across AI/ML systems, enterprise software, and government-facing integrations in East Africa. Built a production containerized AI pipeline (FastAPI, Celery, Redis, Docker, Whisper) for child protection organizations; contributed to an AI-powered KMS for a government energy organisation; engineered a multi-tenant corporate travel SaaS; and delivered enterprise billing and ERP integrations. Track record working with UNICEF, government ministries, and social-impact institutions. Advancing toward AI Product Leadership.

WORK EXPERIENCE

Software Engineer

2025 — Present

BITZ IT Consulting Ltd. · Nairobi, Kenya

Building and maintaining OpenCHS, an AI-powered open-source helpline and case management platform deployed across East Africa. Working across AI integration, backend engineering, system integrations with government and institutional stakeholders, production deployments, and product documentation.

- Built production containerized AI pipeline (FastAPI, Celery, Redis, Docker, Whisper Large V3, spaCy) for audio → transcription → translation → NLP analysis, used by child protection organizations
- Led OpenCHS integrations across Uganda MGLSD, Kenya CPIMS, WhatsApp, and SMS — deployed in 5 countries: Tanzania, Uganda, Kenya, Lesotho, Somalia
- Worked directly with UNICEF Tanzania, UNICEF Kenya, Uganda MGLSD, C-Sema, and CPIMS Kenya as key stakeholders

Python FastAPI Django Java Spring Boot Vue 3 TypeScript Celery Redis Docker Whisper NLP REST APIs PostgreSQL

Software Engineer (Intern → Full-time)

2023 — 2024

Ruel Software · Kenya

Joined as Software Support & Implementation Specialist Intern, supporting Ruel Enterprise ERP implementation and onboarding across 700+ client organizations in 8 sectors. Progressed to full-time Software Engineer, taking on enterprise integration and compliance work.

- Progressed from Software Support & Implementation Specialist Intern to full-time Software Engineer within the same engagement
- Led KRA eTIMS integration for Kenya Revenue Authority — tax compliance across 700+ enterprise clients

Delivered technical support, troubleshooting, and business process training across 8 industry sectors

ERP Systems | Enterprise Software | REST APIs | SQL | Technical Support | Integration

EDUCATION

Bachelor of Science — Computer Science 2019 – 2023

Kabarak University, Nakuru, Kenya

TECHNICAL SKILLS

LANGUAGES

Python, JavaScript, TypeScript, Java, PHP, C#, R

BACKEND

Django, FastAPI, Spring Boot, Node.js, Laravel

FRONTEND

Vue 3, React, Next.js, Vite

AI / ML

Whisper ASR, HuggingFace Transformers, NLLB (Translation), FLAN-T5 / Mistral, spaCy / NLP, Celery + Redis (ML queuing), TensorFlow / PyTorch, Data Science / Jupyter

APIS & INTEGRATIONS

REST APIs, OAuth2 / JWT, RabbitMQ, Swagger / OpenAPI, SSL/TLS & API Security, Postman

TOOLS & DEVOPS

Git / GitHub, Docker, Linux / Ubuntu / WSL, Label Studio, VS Code, Jupyter / Conda

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Digital Product Management: Modern Fundamentals — University of Virginia (Darden School of Business) · Coursera — In Progress

AI Product Management Specialization — Duke University · Coursera — Planned

KEY PROJECTS

OpenCHS (East Africa): AI-powered helpline & case management platform. Deployed in Tanzania, Uganda, Kenya, Lesotho. Clients include UNICEF Tanzania, UNICEF Kenya, Uganda MGLSD, C-Sema, CPIMS Kenya.

Uganda Ministry of Gender Integration: Django-based backend integration with the Uganda MGLSD gateway for child protection data exchange.

Kenya CPIMS Integration: Interoperability integration between OpenCHS and Kenya's national child protection registry (CPIMS).

Tanzania AI Capabilities: Speech-to-text (Whisper), translation (NLLB), summarization (FLAN-T5/Mistral) and classification for AI-enabled case processing.

KRA eTIMS Integration (Ruel Software): Enterprise ERP integration with Kenya Revenue Authority for tax invoice compliance.